

AZHAD SHAHZAD SHAIK

Engineer & Applied Researcher | Security, Machine Learning & Intelligent Systems

New Haven, CT | +1 959-759-3928 | shaikazhadshahzad@gmail.com | github.com/s-shahzad | azhadshahzadshaik.netlify.app

SUMMARY

Engineer and first-author IEEE researcher working at the intersection of security, machine learning, and data systems. M.S. in Cyber/Computer Forensics (Sacred Heart, GPA 3.667). Builds and open-sources production systems (a hybrid ML intrusion-detection platform and a Windows SIEM control plane) and ships healthcare data pipelines in production. Drawn to hard problems in emerging technology: applied AI, critical-infrastructure security, and autonomous systems.

RESEARCH INTERESTS

Applied machine learning for security and anomaly detection · critical-infrastructure and IoT/OT security (EV charging, OCPP) · intrusion detection and behavioral analytics · secure, autonomous data and AI systems · ML-driven automation and agentic tooling.

TECHNICAL SKILLS

Detection & Response: SIEM, EDR, IDS/IPS, MITRE ATT&CK, threat hunting, incident response, alert triage, Sigma rules, anomaly detection, behavioral analytics, log analysis, network forensics, threat intel, false-positive reduction; Wireshark, Nmap, scapy, Nuclei, Gitleaks, custom IDS/SIEM pipelines

ML / Data: scikit-learn (RandomForest, IsolationForest, XGBoost, SVM-RBF, Decision Tree), pandas, SMOTE class balancing, feature engineering for network flow data, ROC/AUROC optimization

Languages, Infra & Frameworks: Python (deep), PowerShell, SQL, Bash, FastAPI, Docker, Linux, Windows host telemetry, AWS, Azure, TCP/IP; NIST CSF, MITRE ATT&CK, OWASP, SOC 2, ISO 27001, HIPAA / PHI, FHIR R4

SELECTED PROJECTS & RESEARCH

Universal NIDS, Hybrid Intrusion Detection Platform

Python, scikit-learn, scapy, FastAPI · Aug 2024 - Present

- Built 4-modality detection fusion (signature 40% + statistical 20% + supervised 30% + unsupervised 10%) using RandomForest and IsolationForest; detects beaconing, exfiltration, lateral movement, and multi-stage campaign chains.
- Production-validated via 6-hour soak test: 87,533 network flows, 0 false alerts at peak 107% CPU / 409 MB RSS; 195 pytest items, 79.16% coverage above a 72% enforced floor.

Vector, Windows Blue-Team SIEM Control Plane

Python, FastAPI, multi-LLM router · Aug 2024 - Present

- Engineered a ~84-file FastAPI control plane fusing Windows host telemetry with NIDS network alerts; SIEM-style correlation produces scored incidents (low / medium / high) with confidence ratings, evidence highlights, and timelines.
- Built AIProviderRouter routing analyst tasks across Ollama, Gemini, OpenAI, and Grok with automatic fallback and per-task attribution; hardened middleware (LocalOnly, API key auth, 120 req/min rate limit) and cut telemetry latency from 3,411 to 2,500 ms. Threat-intel ingestion from stamparm/ipsum (frequency 3+ filter, 6h TTL cache).

IEEE GCAIoT 2025, ML-Based DoS Detection on EV Charging (First author)

Published 2025

- Trained 4 ML classifiers (RandomForest best: 99.03% acc, 99.81% AUROC, 88.83% F1) on a 1.62M-sample CICIDS-derived dataset (SMOTE-balanced); modeled OCPP-protocol attack surface under DoS threat models. DOI: 10.1109/GCAIoT68269.2025.11275540.

PROFESSIONAL EXPERIENCE

Solstice Solutions, Inc. · Jr. Data Analyst (Healthcare PHI Integration)

New Haven, CT · Oct 2025 - Present

- Build and maintain PHI-aware HL7 v2 to FHIR R4 integration pipelines on Azure (ingest, transform, export) covering Medicaid and Medicare Part A/B/C; pipelines hold a 99.9% claims approval rate.
- Process EDI 837 claims and 835 remittance via SQL ETL with FHIR conformance plus HIPAA-aligned validation gates and audit-trail preservation; build Power BI dashboards for pipeline-health and compliance reporting.

Infosysnow Inc. · Cyber Security Engineer

Remote · Aug 2025 - Oct 2025

- Operated network hardening, anomaly triage, and remediation streams using OSI-layer analysis; conducted vulnerability assessments and supported penetration testing.

Sacred Heart University · M.S. Cyber Forensics & Counterterrorism, Research

Fairfield, CT · Jan 2024 - Mar 2025 · GPA 3.667/4.0

- First-author capstone publication at IEEE GCAIoT 2025: ML-based DoS detection on OCPP IoT EV charging infrastructure (critical-infrastructure protection).
- Designed, built, and open-sourced Universal NIDS (hybrid IDS, 79% coverage, soak-validated) and Vector (Windows SIEM control plane) as independent research outputs during the program.

iQ4 · IT Risk Analyst Intern

Remote · May 2024 - Aug 2024

- Led a 350-hour NIST CSF analysis of the Uber 2016 breach (AWS S3 + GitHub credential compromise, 57M users); mapped control failures across all 5 CSF functions, flagged gaps in MFA/RBAC/credential management, delivered a 15-slide remediation briefing.

Sprinto Technology Pvt Ltd · Associate Software Engineer

India · Apr 2023 - Dec 2023

- Backend feature development on a compliance-focused SaaS (SOC 2 / ISO 27001 family); debugged, tested, and validated security-control components for stable releases.

Vodafone Idea Ltd. · Network Engineer (Operations)

Hyderabad, India · Nov 2021 - Apr 2023

- Led OSS-to-SRAN migration for 3,500+ sites across 2G/3G/4G in 15 states; second-level BSC fault troubleshooting.

EDUCATION & CERTIFICATIONS B.Tech ECE, KL University (May 2021, CGPA 8.21) · CompTIA Security+ (SY0-701) · Fortinet NSE 1, 2, 4